

El-Gamal Encryption (1985 — Taher ElGamal)

Core Idea

El-Gamal is an **asymmetric encryption** scheme built on the **Discrete Logarithm Problem**: given (g, p) , and $(g^s \bmod p)$, it is computationally infeasible to find (s) . This is what makes it secure.

Key Generation (Bob)

Step	Operation	Description
1	Choose prime p	Large public prime
2	Choose integer a	$0 \leq a \leq p-1$, public base
3	Choose secret s	$2 \leq s \leq p-2$, kept private
4	Compute $P = a^s \bmod p$	Public value derived from secret

- **Public key** $\rightarrow (p, a, P)$
- **Private key** $\rightarrow s$

Encryption (Alice sends M to Bob)

Alice wants to send message M where $M < p$.

Step	Formula	Description
1	Choose random k	$0 \leq k \leq p-1$, fresh each time
2	$C_1 = a^k \bmod p$	Ephemeral public value
3	$C_2 = M \times P^k \bmod p$	Message masked by shared secret

Alice sends the pair (C_1, C_2) to Bob.

The random k is crucial — it is chosen **fresh for every message**, which is why El-Gamal is called a **probabilistic** cipher (same message M encrypted twice gives different ciphertexts).

Decryption (Bob recovers M)

Bob uses his private key s:

Step	Formula	Result
1	$R = C_1^s \bmod p$	$= (a^k)^s \bmod p = a^{ks} \bmod p = P^k \bmod p$
2	$M = C_2 \times R^{-1} \bmod p$	$= M \times P^k \times P^{-k} \bmod p = M$

Why does $R = P^k \bmod p$?

$$\begin{aligned} R &= C_1^s \bmod p \\ &= (a^k)^s \bmod p \\ &= a^{ks} \bmod p \\ &= (a^s)^k \bmod p \\ &= P^k \bmod p \quad \checkmark \end{aligned}$$

So Bob reconstructs exactly the same P^k that Alice used, and cancels it out.

R^{-1} — Modular Inverse

$R^{-1} \bmod p$ is the number such that $R \times R^{-1} \equiv 1 \bmod p$.

It can be computed using **Fermat's little theorem** (since p is prime):

$$R^{-1} \bmod p = R^{(p-2)} \bmod p$$

So the full decryption formula becomes:

$$M = C_2 \times C_1^{(p-1-s)} \bmod p$$

Because:

$$C_1^{(p-1-s)} = C_1^{(p-1)} \times C_1^{(-s)} = 1 \times C_1^{(-s)} = (a^k)^{(-s)} = P^{(-k)} = R^{-1}$$

Full Example

Parameters: $p = 2579$, $a = 2$, $s = 765$

Key generation:

$$P = a^s \bmod p = 2^{765} \bmod 2579 = 949$$

Public key $\rightarrow (2579, 2, 949)$

Private key $\rightarrow 765$

Alice encrypts $M = 1299$, chooses $k = 853$:

$$C_1 = 2^{853} \bmod 2579 = 435$$

$$C_2 = 1299 \times 949^{853} \bmod 2579 = 2396$$

Ciphertext sent $\rightarrow (435, 2396)$

Bob decrypts:

$$R = C_1^s \bmod p = 435^{765} \bmod 2579 = 949^{853} \bmod 2579$$

$$R^{-1} = R^{(p-2)} \bmod p = R^{2577} \bmod 2579$$

$$M = C_2 \times R^{-1} \bmod p$$

$$= 2396 \times R^{-1} \bmod 2579$$

$$= 1299 \quad \checkmark$$

Summary Table — El-Gamal vs RSA

Feature	RSA	El-Gamal
Hard problem	Integer factorization	Discrete logarithm
Key size	$(e, n) / (d, n)$	$(p, a, P) / s$
Ciphertext size	Same as plaintext	2× plaintext (C_1, C_2)
Deterministic?	Yes	No (random k each time)
Same $M \rightarrow$ same C ?	Yes	No (probabilistic)
Invented	1977	1985